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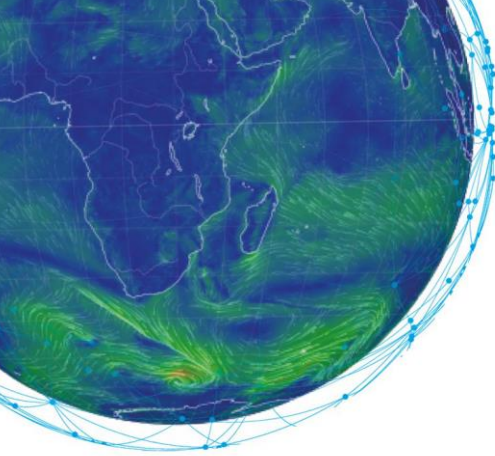
MOOC

Machine Learning in Weather & Climate

TRANSCRIPT

**WeatherBench and the importance
of benchmark datasets**

Expert: Mariana Clare



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WeatherBench and the importance of benchmark datasets

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Introduction

Hello and welcome back. For those who don't know me yet, my name is Lisa Burke and today we'll meet Mariana Clare. Mariana is a mathematician by training but during her career has focussed on applying mathematical techniques to real world Earth Science problems. She is currently a scientist at ECMWF in Germany, where she is using machine learning techniques to quantify uncertainty in high resolution weather forecasts.

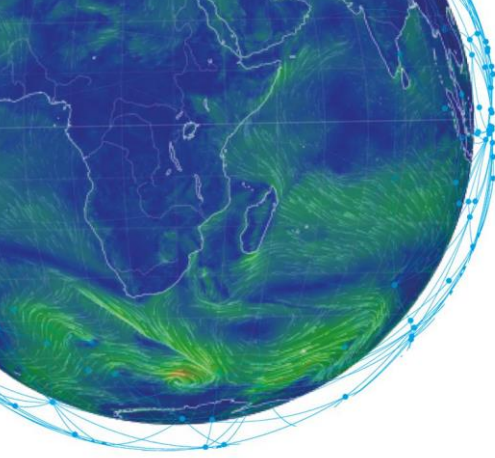
Ready? So, let's go to ECMWF headquarters in Bonn, Germany.

Script

Hello and welcome to the next section of the course. In this section, we will explore the WeatherBench dataset, a dataset designed to be used as a benchmark for data-driven weather forecasting using weather data from ECMWF. This dataset was developed by Stephan Rasp, who has also helped prepare this session in the MOOC. Like any benchmark, WeatherBench aims to support research in the field and provide a trusted evaluation method to enable stable comparisons.

Before we explore the WeatherBench dataset, we want to highlight the importance of benchmark datasets for data-driven problems.

Consider the example problem of training a model to classify images into "dog" or "not dog". Suppose we train two models - model 1 has accuracy 90% and model 2, 70%. Based on these accuracy measures it seems logical that model 1 has done a better job. However the dataset used for Model 1 is predominately images of dogs, whereas we see in the bottom row that the dataset for model 2 includes images of animals. This is just one example of why benchmark datasets are important. Others include that they provide a clear goal for research to aim for and that they provide data and code in an easily accessible format which means that all researchers have to do is download it. Both of these can help speed up progress in research.



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A benchmark dataset is defined by several different properties. Firstly, a benchmark dataset should have a clear problem statement and a good one will be relevant to the community. The dataset and accompanying code should be open source (i.e. freely and readily available to make evaluation reproducible and easy) and the quantitative metrics used to evaluate model accuracy should be unambiguous and appropriate for the problem. Benchmark datasets also contain meaningful baselines to compare against including reference machine learning solutions and simpler computational benchmarks. Finally they should include code for visualisations, diagnostics and robustness tests. Throughout this module, we will show how WeatherBench meets these criteria.

Summary

In this video you have seen a simple example of why benchmark datasets are crucial when you are comparing the accuracy of two different models. You have also learnt the key properties that define a good benchmark dataset. In the rest of the module, you will learn about specific benchmark datasets and understand how they meet these criteria.